

Activity 2: Critical Reflection and Redesign Under Uncertainty and Ethical Constraints

Control Systems Course - 2026-1
Electronic Engineering
Faculty of Engineering
University of Antioquia
Prof. Hernán Felipe García Arias

May 2026

1 Reflecting and Redesigning Control Systems: From Assumptions to Robust and Ethical Design

1.1 Course Context

This activity builds on Activity 1, where students analyzed their two-link walker system from a sociotechnical perspective. In Activity 2, students move beyond anticipation into **critical reflection and redesign**.

Students are now required to **challenge their own assumptions**, explicitly incorporate **uncertainty and real-world variability**, and redesign their controller considering **safety, usability, and ethical constraints**.

This aligns with real-world engineering practice, where systems must operate under uncertainty and interact safely with humans.

1.2 Motivation

In most introductory control courses, systems are designed under ideal assumptions:

- Perfect model knowledge
- Noise-free measurements
- Fully known parameters
- Controlled laboratory conditions

However, in real applications such as rehabilitation systems:

- Users vary significantly (mass, strength, motion patterns)
- Sensors are noisy and imperfect
- Dynamics are uncertain and time-varying
- Safety is critical due to human interaction

Therefore, engineering design must move from:

“Design for performance” → “Design for uncertainty, safety, and real-world use”

This activity introduces the **Reflect** stage of Responsible Innovation:

Reflect = Question your assumptions and evaluate the implications of your design decisions.

1.3 Learning Outcomes

By the end of this activity, students will be able to:

1. Identify and critically evaluate assumptions in their control design.
2. Model uncertainty and disturbances in a control system.
3. Analyze how uncertainty affects stability and performance.
4. Redesign a controller incorporating robustness and safety constraints.
5. Justify engineering decisions considering ethical and real-world implications.

Alignment:

- Design under uncertainty and safety constraints.
- Design → Evaluate → Improve.
- Analyze → Evaluate → Create.

1.4 Core Concept: Reflection in Responsible Innovation

Reflection in engineering means:

- Identifying hidden assumptions in your model
- Questioning whether those assumptions hold in reality
- Evaluating risks introduced by those assumptions
- Redesigning the system accordingly

Students will use guiding questions such as:

- What assumptions did I make about the system?
- What happens if those assumptions are violated?
- Is my system safe under worst-case conditions?
- How does uncertainty affect my controller?

1.5 Technical Framework: Control Under Uncertainty

The system dynamics are now extended to include uncertainty:

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) = \tau + d(t),$$

where:

- $d(t)$ represents disturbances and unmodeled dynamics
- Parameters in $M(q), C(q, \dot{q}), G(q)$ may be uncertain

Linearized uncertain model:

$$\dot{x} = (A + \Delta A)x + (B + \Delta B)u + w(t),$$

where:

- $\Delta A, \Delta B$ represent parametric uncertainty
- $w(t)$ is an external disturbance

Tracking error:

$$e(t) = q_{\text{ref}}(t) - q(t)$$

Performance objective:

$$\lim_{t \rightarrow \infty} e(t) = 0$$

However, under uncertainty, we aim for:

$$\|e(t)\| \leq \epsilon \quad (\text{bounded error})$$

Safety constraints:

$$|\tau_i(t)| \leq \tau_{i,\text{max}}, \quad |\theta_i(t)| \leq \theta_{i,\text{max}}$$

Robust control objective:

$$\min_u \max_{\Delta, w} J$$

where:

$$J = \int_0^T (e^T Q e + u^T R u) dt$$

This formalizes the idea that:

A good controller must perform well **even when the model is not perfect**.

1.6 Learning Activity

1.6.1 Step 1: Identify Design Assumptions (25%)

List and explain at least **five assumptions** made in your original design.

Examples:

- Constant user mass
- Perfect state measurement
- No external disturbances
- Ideal actuator behavior

For each assumption, explain:

- Why it was made
- Whether it is realistic

1.6.2 Step 2: Analyze Impact of Uncertainty (25%)

For each assumption:

- What happens if it is violated?
- How does it affect stability or performance?

Students should connect uncertainty to control performance:

$$x(t) \rightarrow x(t) + \delta x(t) \quad \Rightarrow \quad e(t) \rightarrow e(t) + \delta e(t)$$

1.6.3 Step 3: Safety and Ethical Reflection (20%)

Answer:

- Could your system harm a user under failure?
- What is the worst-case scenario?
- Are safety limits explicitly included?
- Who is responsible if the system fails?

1.6.4 Step 4: Controller Redesign (30%)

Propose a redesigned controller incorporating:

- Robustness (e.g., gain margins, disturbance rejection)
- Safety constraints (e.g., saturation, limits)
- Improved sensing or filtering

Examples:

$$u = -Kx \quad \rightarrow \quad u = -Kx + u_{\text{robust}}$$

or

$$u = \text{sat}(-Kx)$$

Each redesign must include:

- Problem identified
- Control modification
- Technical justification

1.7 Deliverable

Maximum **2 pages** including:

1. Assumptions
2. Uncertainty analysis
3. Safety reflection
4. Controller redesign

1.8 Assessment Rubric

1.9 Expected Learning Closure

At the end of this activity, students should understand:

A controller designed under ideal assumptions is not sufficient. Real engineering requires designing for uncertainty, safety, and human interaction.

Table 1: Assessment rubric for Activity 2.

Criterion	Low	Medium	High
Assumptions	Few or unclear	Some identified	Clear and well-justified
Uncertainty analysis	Descriptive only	Partial reasoning	Strong technical analysis
Safety reflection	Superficial	Some awareness	Deep and critical evaluation
Controller redesign	Weak or missing	Partially justified	Technically sound and well-justified